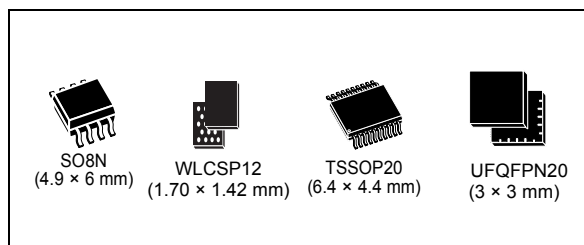


Arm® Cortex®-M0+ 32-bit MCU, 32 KB flash, 6 KB RAM,  
2 x USART, timers, ADC, comm. I/Fs, 2-3.6 V

Datasheet - production data

## Features

- Includes ST state-of-the-art patented technology
- Core: Arm® 32-bit Cortex®-M0+ CPU, frequency up to 48 MHz
- -40°C to 85°C/105°C/125°C operating temperature
- Memories
  - Up to 32 Kbytes of flash memory with protection and securable area
  - 6 Kbytes of SRAM with hardware parity check
- CRC calculation unit
- Reset and power management
  - Voltage range: 2.0 V to 3.6 V
  - Power-on / power-down reset (POR/PDR)
  - Programmable brownout reset (BOR)
  - Low-power modes: Sleep, Stop, Standby, Shutdown
- Clock management
  - 4 to 48 MHz crystal oscillator
  - 32 kHz crystal oscillator with calibration
  - Internal 48 MHz RC oscillator ( $\pm 1\%$ )
  - Internal 32 kHz RC oscillator ( $\pm 5\%$ )
- Up to 18 fast I/Os
  - All mappable on external interrupt vectors
  - All 5 V-tolerant
- 3-channel DMA controller with flexible mapping
- 12-bit, 0.4  $\mu$ s ADC (up to 13 ext. channels)
  - Conversion range: 0 to 3.6 V
- 8 timers: 16-bit for advanced motor control, four 16-bit general-purpose, two watchdogs, SysTick timer
- Calendar RTC with alarm



- Communication interfaces
  - One I<sup>2</sup>C-bus interface supporting Fast-mode Plus (1 Mbit/s) with extra current sink; supporting SMBus/PMBus™ and wake-up from Stop mode
  - Two USARTs with master/slave synchronous SPI; one supporting ISO7816 interface, LIN, IrDA capability, auto baud rate detection and wake-up feature
  - One SPI (24 Mbit/s) with 4- to 16-bit programmable bitframe, multiplexed with I<sup>2</sup>S interface; two extra SPIs through USARTs
- Development support: serial wire debug (SWD)
- SESIP3 and PSA Level 3 target certification
- All packages ECOPACK 2 compliant

Table 1. Device summary

| Reference   | Part number                           |
|-------------|---------------------------------------|
| STM32C011x4 | STM32C011F4, STM32C011J4              |
| STM32C011x6 | STM32C011F6, STM32C011J6, STM32C011D6 |